📐 Geometry Handbook — Formulas & Figures (SSC CGL)

> Every shape & theorem with a **labelled figure** + its formulas. Companion to SSC-CGL-ASO-Formulas-Tricks-Master.md. Syllabus: Quant — Geometry & Mensuration (Para 13.10.3 / 13.11.1.4–13.11.1.5). View in a monospace font (or the HTML/PDF) so the figures stay aligned.

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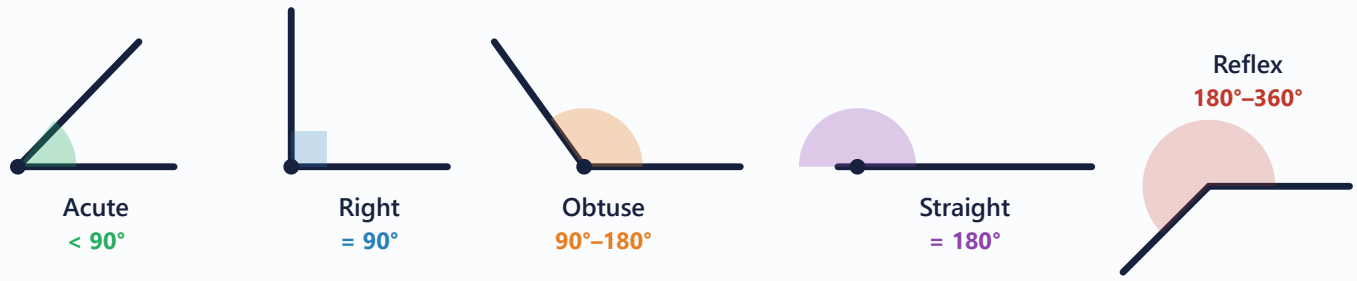
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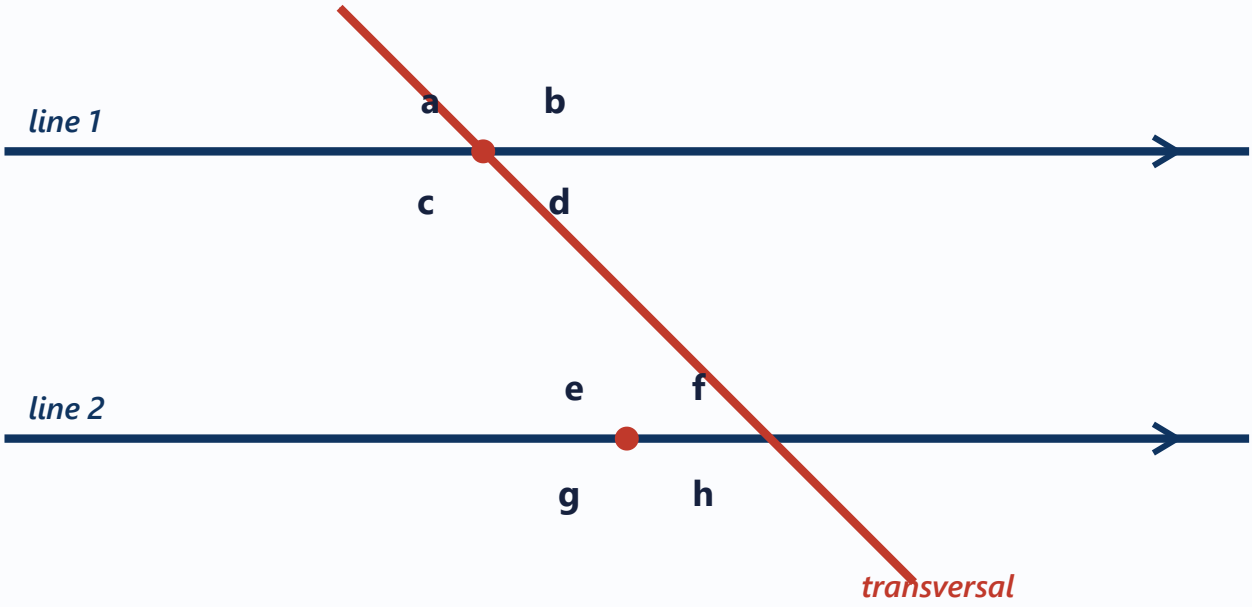
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1. Lines & Angles

① Types of angles

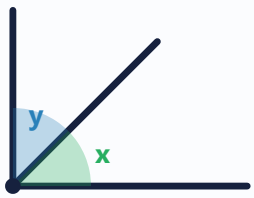


② A transversal cutting two parallel lines

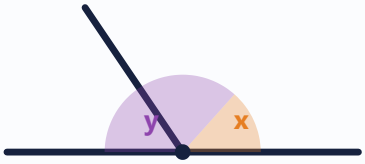


Pair (in figure)	Relationship
Corresponding — **a&e, b&f, c&g, d&h**	**equal**
Alternate interior — **c&f, d&e**	**equal**
Alternate exterior — **a&h, b&g**	**equal**
Co-interior (allied) — **c&e, d&f**	sum = **180°**
Vertically opposite — **a&d, b&c** (and e&h, f&g)	**equal**

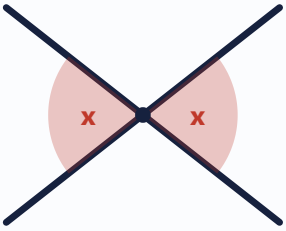
③ Angle-pair definitions



Complementary
x + y = 90°



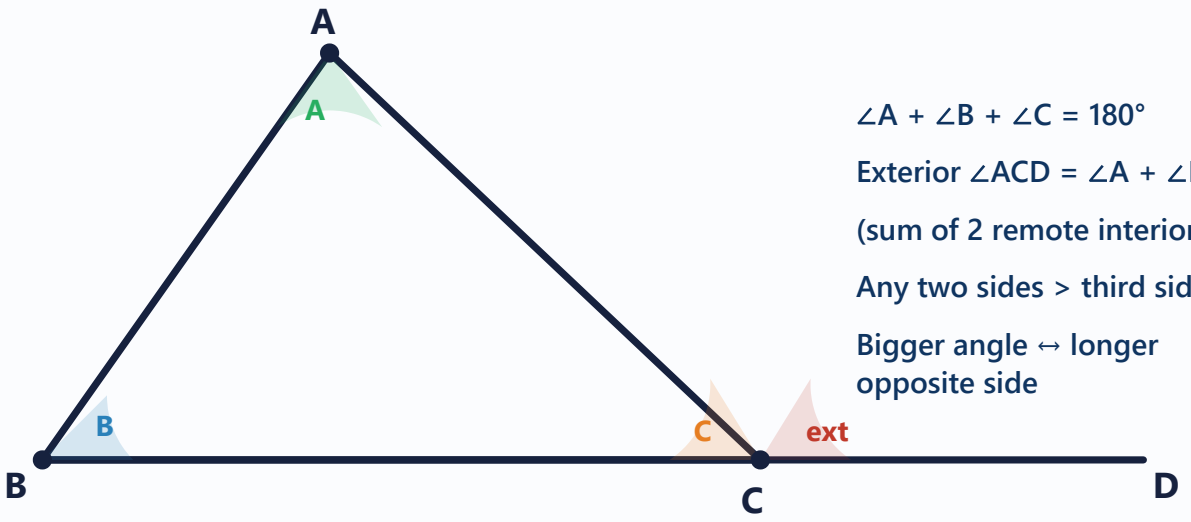
Supplementary
x + y = 180°



Vertically opposite
the two x angles are equal

Concept	Rule
Complementary angles	sum = **90°**
Supplementary angles	sum = **180°**
Angles on a straight line	sum = **180°** (linear pair)
Angles around a point	sum = **360°**
Vertically opposite angles	**equal** (a = d, b = c)
Parallel lines: corresponding	**equal**
Parallel lines: alternate (interior/exterior)	**equal**
Parallel lines: co-interior (allied)	sum = **180°**

2. Triangle — basics & types



∠A + ∠B + ∠C = 180°

Exterior ∠ACD = ∠A + ∠B

(sum of 2 remote interiors)

Any two sides > third side

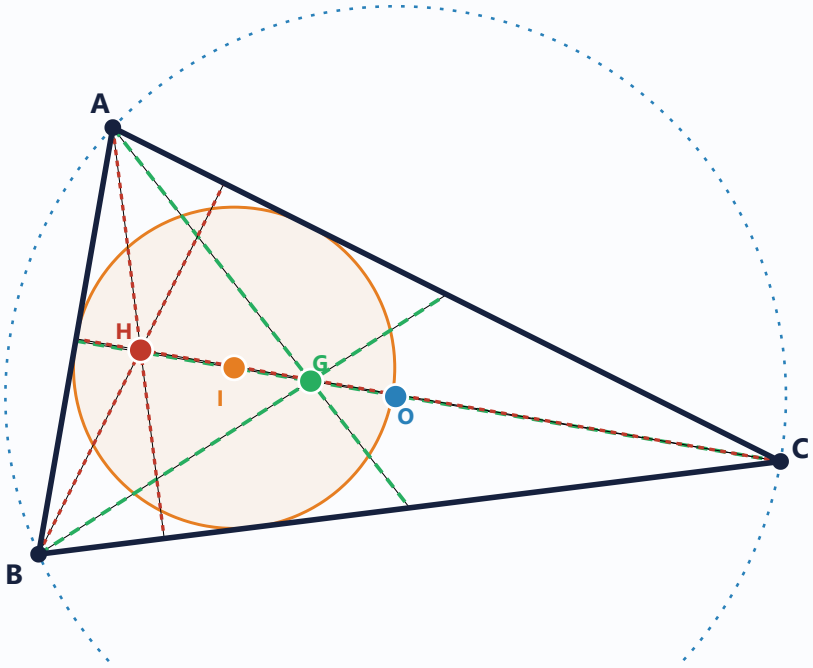
Bigger angle ↔ longer opposite side

By sides	By angles
Equilateral — all sides equal (each angle 60°)	**Acute** — all angles < 90°
Isosceles — two sides equal (base angles equal)	**Right** — one angle = 90°
Scalene — all sides different	**Obtuse** — one angle > 90°

3. The Four Centres of a Triangle

Centre	Where lines meet	Note
Centroid G	the 3 **medians** (vertex → midpoint)	divides each median **2:1** (from vertex)
Incentre I	the 3 **angle bisectors**	centre of **inscribed** circle
Circumcentre O	the 3 **perp. bisectors** of the sides	centre of **circumscribed** ○; equidistant from vertices
Orthocentre H	the 3 **altitudes** (vertex ⊥ opposite side)	—

- G, O, H are collinear → the **Euler line**; and OG : GH = **1 : 2**.
- In an equilateral triangle all four centres **coincide**.



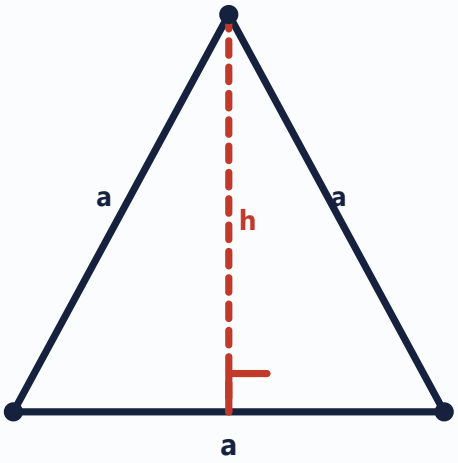
- **G** — Centroid
  3 medians · splits each 2 : 1
- **I** — Incentre
  3 angle bisectors · inscribed ○
- **O** — Circumcentre
  3 perp. bisectors · circumscribed ○
- **H** — Orthocentre
  3 altitudes (vertex ⊥ opp. side)

Euler line: O, G, H are collinear, with OG : GH = 1 : 2.
Equilateral △ → all four coincide.

Radii: inradius **r = Area / s** · circumradius **R = (a·b·c) / (4·Area)**.
In a **right triangle**, R = hypotenuse / 2 and the circumcentre O is the **midpoint of the hypotenuse**.

4. Triangle Area Formulas

Given	Area
base & height	**½ × base × height**
3 sides a,b,c (Heron)	**√[s(s−a)(s−b)(s−c)]**, s = (a+b+c)/2
two sides & included angle	**½ · a · b · sin C**
equilateral side a	**(√3 / 4) · a^2**
with inradius r	**r × s**
with circumradius R	**(a·b·c) / (4R)**



height $h = (\sqrt{3} / 2) \cdot a$

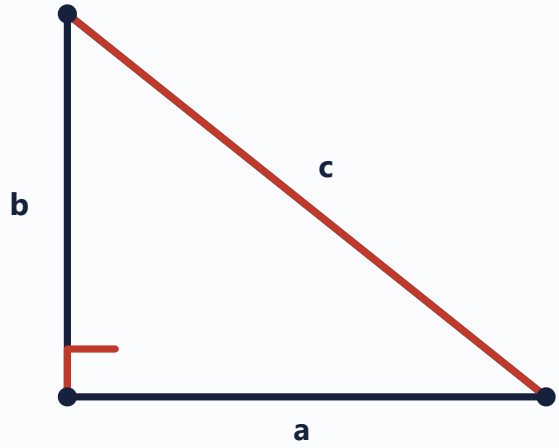
area $= (\sqrt{3} / 4) \cdot a^2$

inradius $r = a / (2\sqrt{3})$

circumradius $R = a / \sqrt{3}$

(so $R = 2r$)

5. Pythagoras Theorem & Triples



$c^2 = a^2 + b^2$

hypotenuse^2 = sum of squares of legs

Common Pythagorean triples:

3, 4, 5 5, 12, 13

8, 15, 17 7, 24, 25

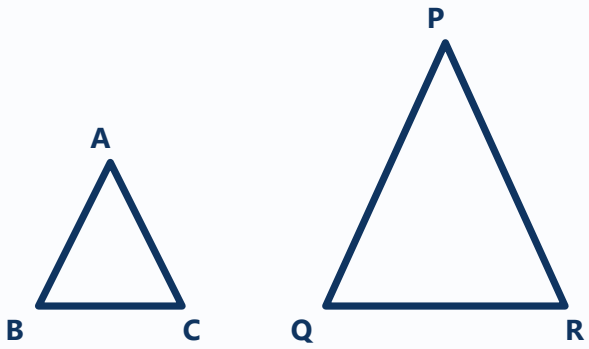
9, 40, 41 20, 21, 29

...and any multiple (6-8-10, 9-12-15)

Result	Formula
Hypotenuse	$\sqrt{(a^2 + b^2)}$
A leg	$\sqrt{(c^2 - \text{other leg}^2)}$
Altitude to hypotenuse	$(a \cdot b)/c$
Right-triangle inradius	$r = (a + b - c)/2$

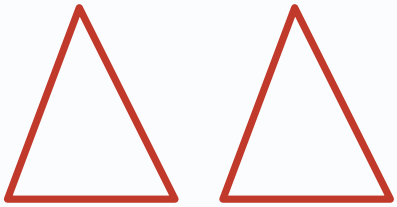
6. Similarity & Congruence

SIMILAR — same shape, scaled



AB/PQ = BC/QR = CA/RP = k

CONGRUENT — identical



Congruence tests: SSS, SAS, ASA, AAS, RHS

Similarity tests: AA, SSS~, SAS~ (RHS = right △

In similar triangles (ratio k)	Relationship
Corresponding sides	in ratio **k**
Perimeters	in ratio **k**
Areas	in ratio $\mathbf{k^2}$
(for solids) Volumes	in ratio $\mathbf{k^3}$
Basic Proportionality (Thales)	a line ∥ one side divides the other two **proportionally**

7. Quadrilaterals

Square

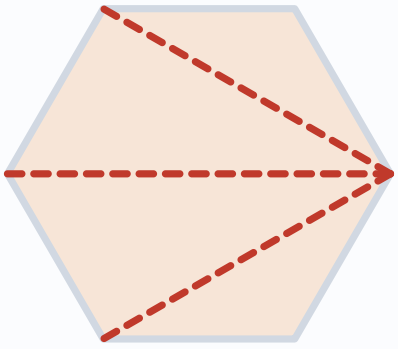
Rectangle

Parallelogram

Rhombus

Shape	Area	Perimeter / notes
Square (side a)	a^2	4a; diagonal = $a\sqrt{2}$
Rectangle (l, b)	l·b	$2(l+b)$; diagonal = $\sqrt{(l^2+b^2)}$
Parallelogram	base × height	opposite sides ∥ & equal; diagonals bisect
Rhombus (diagonals d_1,d_2)	**$\frac{1}{2} \cdot d_1 \cdot d_2$**	all sides equal; diagonals ⊥ bisect
Trapezium (∥ sides a,b; height h)	**$\frac{1}{2}(a + b) \cdot h$**	one pair of sides parallel
Kite (diagonals d_1,d_2)	$\frac{1}{2} \cdot d_1 \cdot d_2$	diagonals ⊥
Cyclic quad (Brahmagupta)	$\sqrt{[(s-a)(s-b)(s-c)(s-d)]}$	opposite angles sum 180°

8. Polygons



Regular hexagon (n = 6)

Sum of interior angles = (n − 2) × 180°

Each interior (regular) = (n−2)·180° / n

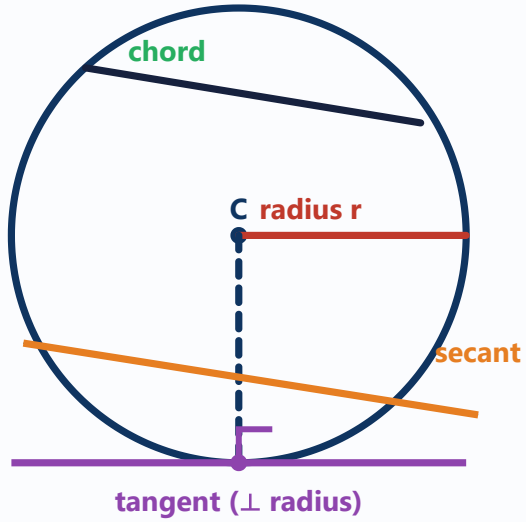
Each exterior (regular) = 360° / n

Number of diagonals = n(n − 3) / 2

Hexagon: interior 120°, 9 diagonals

n	Name	Interior-angle sum	Each interior (regular)
3	Triangle	180°	60°
4	Quadrilateral	360°	90°
5	Pentagon	540°	108°
6	Hexagon	720°	120°
8	Octagon	1080°	135°

9. Circle — parts & theorems



Radius r; Diameter d = 2r

Circumference = 2πr = πd

Area = πr^2

Chord — joins two points on ○

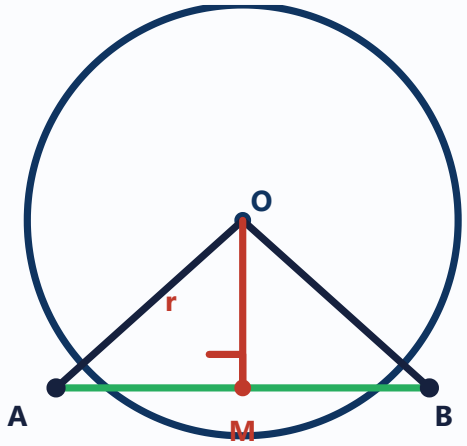
Tangent — touches at 1 point, ⊥ radius

Secant — cuts ○ at two points

Theorem	Statement
Angle at centre	= **2 ×** angle at circumference (same arc)
Angles in same segment	**equal**
Angle in a semicircle	**90°** (diameter subtends right angle)
Cyclic quadrilateral	opposite angles sum **180°**
Tangent ⊥ radius	at the point of contact (90°)
Two tangents from a point	are **equal** in length
Alternate segment	tangent-chord angle = angle in alternate segment
Intersecting chords	PA · PB = PC · PD
Equal chords	equidistant from the centre

Circle measure	Formula
Area	$\pi\ r^2$
Circumference	$2\ \pi\ r$
Arc length (angle θ°)	$(\theta/360) \times 2\pi r$
Sector area (angle θ°)	$(\theta/360) \times \pi r^2$
Segment area	sector area − triangle area

9A. Circle — Chord Theorems



OM ⊥ AB ⇒ AM = MB

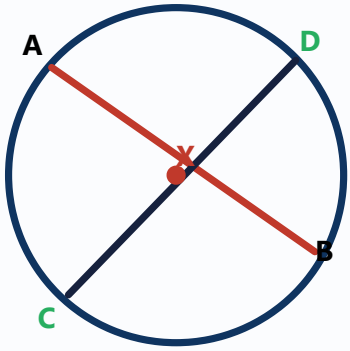
Distance of chord from centre:

$OM = \sqrt{(r^2 - (AB/2)^2)}$

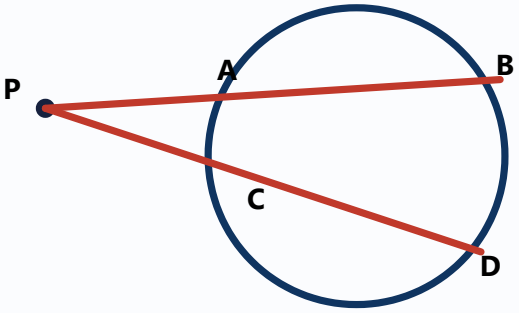
Conversely, the line from O to the midpoint of a chord is ⊥ to it.

Chord theorem	Statement
Perpendicular from centre	**bisects** the chord (AM = MB)
Line from centre to chord-midpoint	is **perpendicular** to the chord
Equal chords	are **equidistant** from the centre (and vice-versa)
Longer chord	is **nearer** the centre; the diameter is the longest chord
Equal chords subtend	**equal angles** at the centre

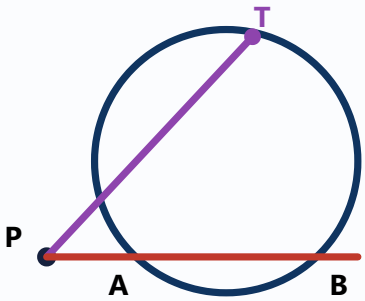
9B. Intersecting Chords, Secants & Tangent–Secant (Power of a Point) ⭐



① **Chords inside**
AX·XB = CX·XD



② **Two secants**
PA·PB = PC·PD

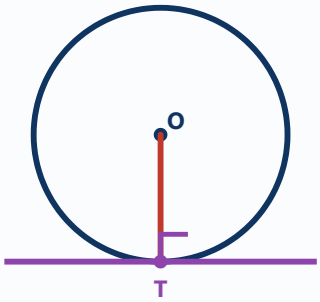


③ **Tangent + secant**
$PT^2 = PA \cdot PB$

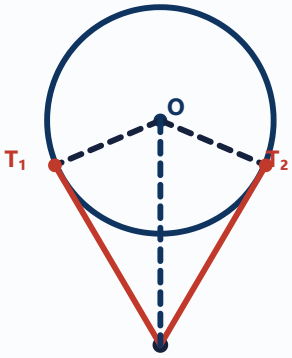
Theorem (Power of a Point)	Relation
Two chords intersect **inside** at X	AX · XB = CX · XD
Two secants from an **external** point P	PA · PB = PC · PD
Tangent–Secant from external point P	$PT^2 = PA \cdot PB$
Two **tangents** from external point P	$PT_1 = PT_2$ (equal lengths)

 One idea: for any point P and a line through it cutting the circle at two points, the **product of the two distances is constant** (the "power" of P). Inside → chord·chord; outside → secant·secant; tangent is the limiting case where the two points merge → PT^2.

9C. Tangent Theorems



Tangent ⊥ radius (90°)



P (external)

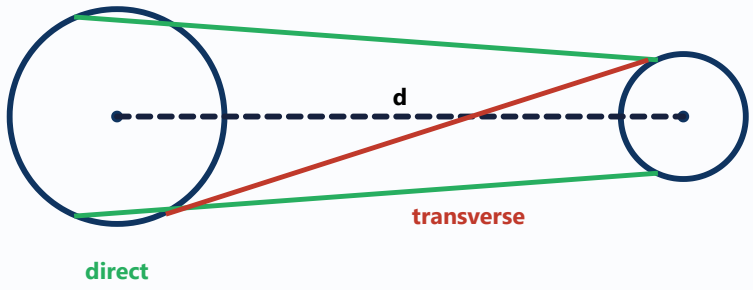
$PT_1 = PT_2$

OP bisects $\angle T_1PT_2$

$OT_1 \perp PT_1$

$OT_2 \perp PT_2$

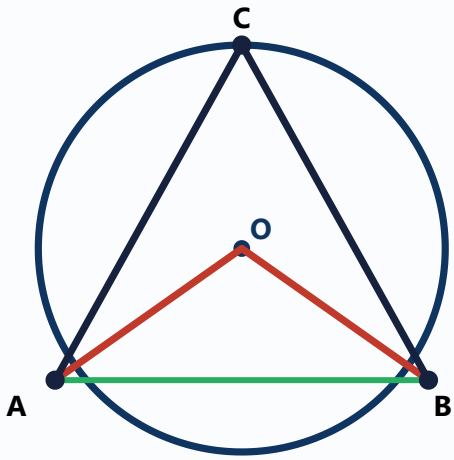
Tangent theorem	Statement
Tangent ⊥ radius	at the point of contact (90°)
Two tangents from external point	are **equal** ($PT_1 = PT_2$)
Line from centre to external point	**bisects** the angle between the two tangents
Tangent length	$PT = \sqrt{(OP^2 - r^2)}$
Alternate Segment Theorem	angle between tangent & chord = angle in the **alternate segment**



Direct common tangent
$= \sqrt{(d^2 - (R - r)^2)}$

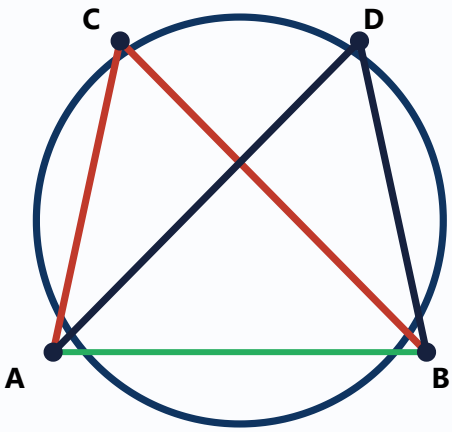
Transverse common tangent
$= \sqrt{(d^2 - (R + r)^2)}$

9D. Proportionality & Angle Theorems (Circle)



Centre angle = 2 × circumference

$\angle AOB = 2 \cdot \angle ACB$ (arc AB)



Angles in same segment

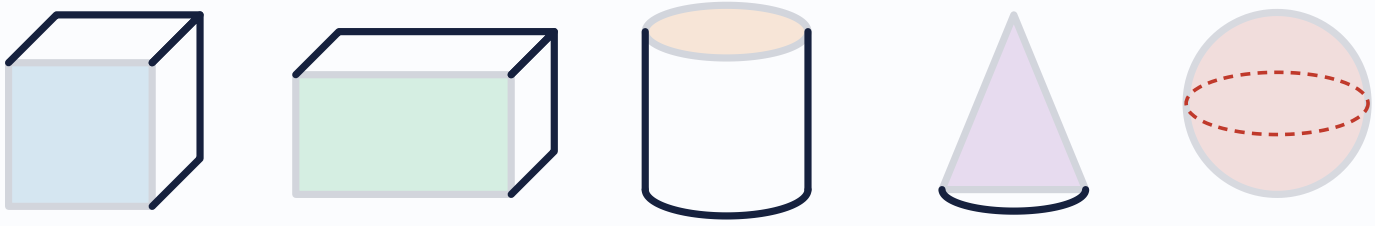
$\angle ACB = \angle ADB$ (arc AB)

Angle / proportion theorem	Statement
Angle at centre	= **2 ×** angle at the circumference on the same arc
Angles in the same segment	are **equal**
Angle in a semicircle	= **90°**
Cyclic quadrilateral	opposite angles are **supplementary** (sum 180°)
Exterior angle of cyclic quad	= interior **opposite** angle
Ptolemy's theorem (cyclic quad)	$AC \cdot BD = AB \cdot CD + AD \cdot BC$ (product of diagonals)
Equal arcs	subtend equal angles at the centre & equal chords

10. Mensuration 2D — areas & perimeters

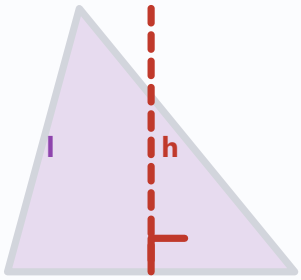
Shape	Area	Perimeter
Square (a)	a^2	4a
Rectangle (l,b)	l·b	2(l+b)
Triangle (b,h)	½·b·h	a+b+c
Equilateral (a)	$(\sqrt{3}/4)a^2$	3a
Parallelogram	b·h	2(a+b)
Rhombus (d_1,d_2)	½·d_1·d_2	4·side
Trapezium (a,b,h)	½(a+b)h	sum of sides
Circle (r)	πr^2	$2\pi r$
Semicircle (r)	$\frac{1}{2}\pi r^2$	$\pi r + 2r$
Ring (R,r)	$\pi(R^2-r^2)$	—

11. Mensuration 3D — solids

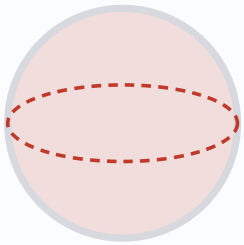


Cube **Cuboid** **Cylinder** **Cone** **Sphere**

Solid	Volume	Curved/Lateral SA	Total SA
Cube (a)	a^3	$4a^2$	$6a^2$; diagonal $a\sqrt{3}$
Cuboid (l,b,h)	$l \cdot b \cdot h$	$2h(l+b)$	$2(lb+bh+hl)$; diag $\sqrt{(l^2+b^2+h^2)}$
Cylinder (r,h)	$\pi r^2 h$	$2\pi rh$	$2\pi r(r+h)$
Cone (r,h)	$\frac{1}{3}\pi r^2 h$	πrl	$\pi r(r+l)$; slant $l=\sqrt{(r^2+h^2)}$
Sphere (r)	$4/3 \cdot \pi r^3$	—	$4\pi r^2$
Hemisphere (r)	$2/3 \cdot \pi r^3$	$2\pi r^2$	$3\pi r^2$
Frustum (R,r,h)	$\frac{1}{3}\pi h(R^2+r^2+Rr)$	$\pi(R+r)l$	$+ \pi(R^2+r^2)$

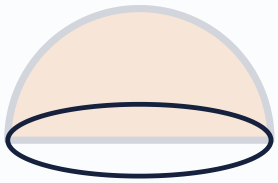


Cone: $l^2 = r^2 + h^2$



$V = (4/3) \cdot \pi r^3$

$SA = 4\pi r^2$

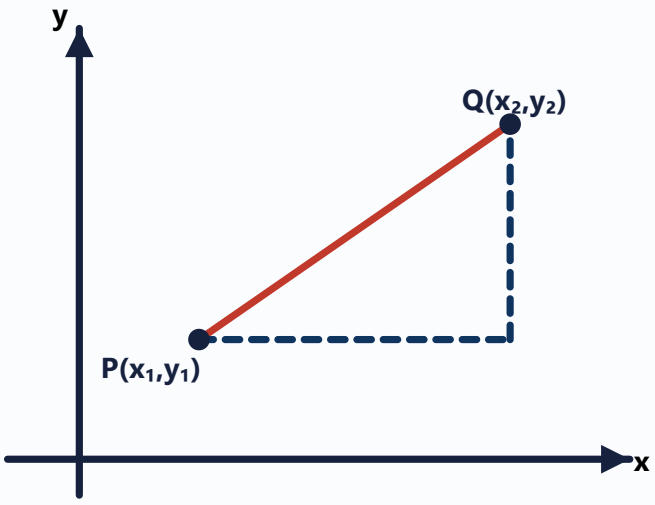


$V = (2/3) \cdot \pi r^3$

$TSA = 3\pi r^2$

(flat base + curved)

12. Coordinate Geometry



Distance = $\sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$

Midpoint = $((x_1+x_2)/2 , (y_1+y_2)/2)$

Slope m = $(y_2-y_1)/(x_2-x_1)$

Section (m:n) =
$((mx_2+nx_1)/(m+n),$
$(my_2+ny_1)/(m+n))$

Concept	Formula
Distance	$\sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$
Midpoint	$((x_1+x_2)/2, (y_1+y_2)/2)$
Slope m	$(y_2-y_1)/(x_2-x_1)$
Line (slope-intercept)	$y = mx + c$
Parallel lines	equal slopes ($m_1 = m_2$)
Perpendicular lines	$m_1 \cdot m_2 = -1$
Area of triangle (vertices)	$\frac{1}{2} \lvert x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2) \rvert$

13. Quick recall table

Prompt	Answer
Triangle angle sum	180°
Equilateral area	$(\sqrt{3}/4)\cdot a^2$
Heron's area	$\sqrt{[s(s-a)(s-b)(s-c)]}$
Centroid divides median	2 : 1
Circumradius R	abc/(4·Area)
Inradius r	Area/s
Angle in semicircle	90°
Angle at centre vs circumference	2 : 1
Rhombus area	$\frac{1}{2}\cdot d_1\cdot d_2$
Trapezium area	$\frac{1}{2}(a+b)\cdot h$
Interior-angle sum (n-gon)	$(n-2)\cdot 180°$
Diagonals of n-gon	$n(n-3)/2$
Cube diagonal	$a\sqrt{3}$
Cuboid diagonal	$\sqrt{(l^2+b^2+h^2)}$
Cone slant l	$\sqrt{(r^2+h^2)}$
Sphere volume / SA	$4/3\cdot\pi r^3$ / $4\pi r^2$
Hemisphere TSA	$3\pi r^2$
Areas of similar figures	ratio of sides^2
Perpendicular slopes	$m_1\cdot m_2 = -1$

✅ **Strategy:** learn the **figure first**, then attach its formulas — recall is far stronger when you can *picture* the shape. Redraw 2–3 figures from memory daily. Back to the main formula sheet ».